

Task 2 — Model Comparison Report

Feature Engineering, Model Optimization & Performance Comparison

Maincrafts Technology | AI & ML Internship

1. Objective

Building on the baseline linear regression from Task 1 ($R^2 \sim 0.60$), this task aims to improve model performance through proper preprocessing and by comparing multiple regression algorithms. The goal is to identify the best-performing model and understand why it outperforms the others.

2. Dataset

The California Housing Dataset (scikit-learn) was used. It contains 20,640 samples across 8 features: MedInc, HouseAge, AveRooms, AveBedrms, Population, AveOccup, Latitude, and Longitude. The target variable is median house value in units of \$100k, capped at \$500k.

3. Methodology

3.1 Feature Scaling

StandardScaler was applied to all features before model training. Without scaling, Population (range: 3-35,682) dominates features like HouseAge (range: 1-52), causing the model to weight them unfairly. After scaling, all features have mean=0 and std=1. The scaler was fit only on training data to prevent data leakage.

3.2 Train/Test Split

An 80/20 split was used with random_state=42 for reproducibility — matching Task 1 to allow direct performance comparison.

3.3 Models Trained

Three regression models were trained and evaluated:

Model	Rationale
Linear Regression	Baseline — same as Task 1 but now with scaled features
Ridge Regression (alpha=1.0)	L2 regularisation; reduces overfitting when features correlate
Decision Tree (max_depth=5)	Captures non-linear patterns; depth capped to avoid overfit

4. Results

All models were evaluated on the same held-out test set using three metrics — RMSE, MAE, and R^2 . Lower RMSE/MAE and higher R^2 indicate better performance.

Model	RMSE	MAE	R2 Score
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Linear Regression	~0.745	~0.531	~0.596
Ridge Regression	~0.745	~0.531	~0.596
Decision Tree (depth=5)	~0.694	~0.494	~0.647

5. Analysis & Discussion

Linear vs Ridge

Linear and Ridge produced nearly identical scores. Ridge's regularisation is most useful with high-dimensional data or severe multicollinearity — neither condition is strongly present here with only 8 features. At $\alpha=1.0$ the penalty is mild enough to have negligible effect.

Decision Tree Improvement

The Decision Tree improved R^2 from ~0.596 to ~0.647 — a meaningful gain. This confirms the presence of non-linear relationships in the data. Location features (Latitude, Longitude) and income likely interact in threshold-based ways that a tree captures well but a linear model cannot.

Effect of Feature Scaling

Scaling had a modest direct effect on linear regression's score but is still best practice. Its biggest impact would be seen with gradient-based optimisers or higher-alpha Ridge/Lasso, where unscaled features cause convergence issues.

6. Conclusion

The Decision Tree Regressor ($\text{max_depth}=5$) was selected as the best model, achieving $R^2 \sim 0.647$ and RMSE ~0.694 (\$69,400 average prediction error). This is a clear improvement over the Task 1 baseline of $R^2 \sim 0.596$.

The logical next step is ensemble methods — Random Forest or Gradient Boosting — which typically reach R^2 0.80+ on this dataset by combining many trees and reducing variance.